

Avranil Dhar

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SUMMARY

- Final Year Engineering student applying C, Java, and JavaScript to develop full-stack web applications. Experienced in building RAG-based AI applications with React, Node.js, Express.js, LangChain, vector databases, and asynchronous processing pipelines. Strong foundation in Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, and Computer Networks, with a focus on building scalable and practical software solutions.

TECHNICAL SKILLS

- ▷ **Languages:** C (Intermediate), Java (Intermediate), JavaScript (Basic), Python (Basic), SQL (Intermediate)
- ▷ **Frontend:** HTML, CSS, React.js, Tailwind CSS, Shadcn/UI
- ▷ **Backend:** Node.js, Bun, Express.js, LangChain.js
- ▷ **Databases:** MySQL, MongoDB, Redis
- ▷ **Tools & Platforms:** Git, GitHub, Docker, Postman, Mongoose, Clerk, Vercel, Render
- ▷ **Computer Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Computer Networks

PROJECTS

Lumina | RAG Document Intelligence System | [GitHub](#) [Live](#) Jun 2026 – Aug 2026

- Built a full-stack RAG platform using **React**, **Node.js**, **Express.js**, and **LangChain** enabling users to upload PDFs and interact with them through context-aware retrieval powered by **Pinecone** and **Groq**.
- Designed an asynchronous document ingestion pipeline using **BullMQ**, **Redis**, and **MongoDB** to decouple document processing from the web server, enabling reliable and scalable document indexing.
- Implemented secure authentication and per-user document scoping using **Clerk**, with the application deployed across **Render** and **Vercel**.

Ionospheric TEC Forecasting & Ionospheric Delay Correction | [GitHub](#) Mar 2026 – Jul 2026

- Led a mentored group project forecasting 24-hour-ahead Total Electron Content (**TEC**) to mitigate ionospheric delays in **GNSS**, using real **NavIC/IRNSS** satellite data from IISc Bangalore.
- Built and benchmarked **LSTM** and **Transformer** models in **PyTorch** for TEC forecasting and derived **L1/L5** band ionospheric delays using thin-shell mapping.
- Evaluated geomagnetic storm detection using **Probability of Detection (POD)**, **Critical Success Index (CSI)**, and **F1-score** alongside forecast error to assess model reliability during solar disturbances.

Codewithavra | Personal Portfolio Website | [GitHub](#) [Live](#) Apr 2026 – May 2026

- Crafted a visually polished portfolio with **React** and **TypeScript**, implementing dark/light mode theming, fluid text animations, and scroll-driven transitions using **Motion** and **Lenis** smooth scroll.
- Engineered a fully responsive, performant UI with **Tailwind CSS v4**, focusing on micro-interactions, typography, and visual hierarchy to deliver a production-grade frontend experience.
- Deployed on **Vercel** with CI/CD via GitHub and integrated **Vercel Analytics** for real-time visitor insights.

CERTIFICATIONS

Programming in Java – NPTEL (IIT Kharagpur) | Ranked Top 1% (99%) | [Certificate](#) Jul – Oct 2025

Full Stack Development with MEAN Stack – Ardent Computech | 36 hours training | [Certificate](#) Jul 2025

Introduction to Database Systems – NPTEL (IIT Madras) | Ranked Top 1% (90%) | [Certificate](#) Jan – Apr 2025

Python with Django – Ardent Computech | 30 hours web development training | [Certificate](#) Feb 2025

ACHIEVEMENTS

HackerRank Badges | 4★ SQL, 3★ JAVA Aug 2026

NIT Jewels | Academic Excellence Award | Narula Institute of Technology Jul 2026

NIT Jewels | Academic Excellence Award | Narula Institute of Technology May 2025

Finalist, Circuithon 2024 | IEEE IEM CAS | Top finalist among 150 participants in circuit design | [Certificate](#) Dec 2024

EDUCATION

B.Tech in Electronics & Communication Engineering | Narula Institute of Technology | **CGPA: 9.5**(upto 5th sem) 2023 – 2027

10+2 (WBCHSE) | Kalyan Nagar Vidyapith | **76.33%** 2022

10th (WBBSE) | Kalyan Nagar Vidyapith | **81.86%** 2020